

Generative AI Interfaces as Cognitive Signals: How UI Structure Shapes Judgment Termination and User Revision

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Abstract

Generative AI interfaces play a role in shaping when users decide to continue or stop evaluating AI-generated artifacts. While prior Human–AI Interaction research has emphasized usability, transparency, and output quality, less attention has been paid to how interface design organizes users’ judgment processes over time. We examine generative AI interfaces through the lens of cognitive signals that shape when users decide to terminate judgment, influencing whether AI-generated outputs are treated as provisional results or judgment-ready artifacts. Using a within-subjects experiment, we compared three interface conditions with Low, Medium, and High levels of cognitive signals while holding the AI model, prompt, and generated output constant. This study examines how the visibility and structural organization of editable points influence users’ interpretation of AI-generated artifacts and, critically, when judgment terminates. Higher levels of cognitive signals were associated with sustained revision and deferred judgment termination, whereas low-signal interfaces more often led to early stopping driven by uncertainty about what to do next. These findings suggest that interface structure relates specifically to when users decide to continue or stop judgment, highlighting judgment timing as a distinct dimension of generative AI interaction even when output quality remains unchanged.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; *Empirical studies in HCI*; Natural language interfaces.

Keywords

generative AI, user interface design, cognitive load, judgment termination, UX evaluation

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1 Introduction

In generative image creation, deciding when to stop interacting with an AI system is inherently ambiguous. The same output may be accepted as complete by one user while treated as provisional by another. Unlike deterministic editing tools, generative AI systems produce probabilistic artifacts without clear completion criteria. Users must therefore judge not only output quality, but whether further interaction is likely to meaningfully improve an uncertain result. This suggests that stopping decisions are shaped not solely by outputs, but by how interfaces render next actions visible and interpretable.

Prior Human–AI Interaction research has articulated guidelines for interactive AI systems [1] and emerging design principles for generative AI applications [11, 12], as well as broader interaction patterns in human–generative-AI systems [10]. However, comparatively less attention has been paid to how interface structure organizes users’ interpretation of AI-generated outputs over time—particularly in relation to when users decide to continue revising or terminate judgment.

We use the term *cognitive signals* to describe interface-level structural cues that make potential next actions perceptible and temporally organized during evaluation. While overlapping with established concepts such as affordances [4, 7], this term is adopted as an analytic lens to foreground judgment timing in open-ended generative workflows. Rather than proposing a new theoretical construct, we examine how the visibility and organization of editable points influence whether AI-generated outputs are interpreted as provisional or complete [9].

To isolate interface effects, we compared three UI conditions—Low, Medium, and High cognitive signal levels—while holding the AI model, prompt, and generated output constant. Across a controlled image-generation task and follow-up interviews, weaker structural cues were associated with earlier stopping, often described as uncertainty about what to do next. More explicitly structured cues supported sustained revision and delayed termination, with stopping framed as a deliberate decision following exploration. These findings suggest that in generative AI contexts—where outputs are probabilistic and stopping criteria ambiguous—interface-level cues play a salient role in shaping judgment termination.



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2 Related Work

2.1 Generative AI Interfaces and User Interpretation

Human–AI Interaction research has articulated guidelines for interactive AI systems [1] and design principles for generative AI applications [11, 12]. Survey work has further characterized emerging interaction patterns in human–generative-AI systems [10].

However, much of this literature emphasizes transparency, prompting strategies, and output evaluation. Comparatively less attention has been paid to how interface structure organizes users’ interpretation of AI-generated outputs over time, particularly in relation to when users decide to continue revising or terminate judgment. In generative contexts, where outputs are probabilistic and completion criteria are ambiguous, such timing decisions become especially salient.

2.2 Affordances and Interface Cues in Judgment

Affordance-based perspectives conceptualize design elements as signaling possible actions [4, 7]. Interface research has further shown that structural cues foreground certain actions while deferring others, thereby shaping user expectations and behavior [5]. Building on this, cognitive modeling in HCI provides a comparative basis for understanding user behavior during interface interaction [6], while specific theories in GUI design emphasize the application of cognitive psychology to improve interaction quality [14].

Building on this work, we focus not only on action possibility but on how the visibility and organization of interface cues relate to judgment timing in open-ended generative workflows. We use the term *cognitive signals* to denote these structural cues as they shape transitions between inspection, revision, and termination decisions.

2.3 Editing and Iteration in Generative Workflows

Recent studies have examined iterative interaction and composable prompting in generative systems [2, 3]. These systems highlight revision and exploration—often through the iterative evolution of prompts to refine outputs [13]—yet the associated usability challenges heavily influence user acceptance and trust [8]. Despite this, stopping decisions are often treated as implicit outcomes of task completion. In contrast, we examine judgment termination directly, isolating interface structure while holding AI output constant to understand how configuration relates to when users decide to stop revising.

3 Methodology

We examined how users interpreted AI-generated outputs and made stopping decisions across interface conditions that varied in the visibility and organization of editable points, while holding the underlying AI output constant.

3.1 Experimental Design and Interface Conditions

Three interface conditions systematically varied the visibility, placement, and immediacy of editable controls while preserving identical

AI-generated output (Fig. 1). The manipulated elements included the number of visible editing controls, their spatial proximity to the output, and whether revision actions were presented as primary or secondary interaction options.

Low Cognitive Signal UI (UI-1). The output was presented prominently with editing controls hidden behind secondary menus. Users could accept or reject the output directly, while revision required navigating to an additional panel. This configuration foregrounded evaluation over iteration.

Medium Cognitive Signal UI (UI-2). A limited subset of editing controls (e.g., regenerate, minor adjustments) was visibly accessible alongside the output. However, advanced editing remained nested, preserving the output as a primary evaluation unit while signaling that revision was possible.

High Cognitive Signal UI (UI-3). Multiple editing controls were directly integrated around the output, including regenerate, parameter adjustments, and iterative refinement options. Revision was presented as the default interaction flow rather than a secondary action.

All participants experienced all three conditions in a randomized within-subjects design.

3.2 Task and Data Collection

Participants completed an open-ended image generation task without objective correctness criteria, requiring them to decide when an output was “good enough” to finalize. All participants used the same prompt (“Generate a visual image to be used for a concept poster”) to isolate interface effects from output variation.

Fourteen participants ($N = 14$) completed the task across all three conditions. After viewing the initial output, participants interacted with the interface as desired, including revising or regenerating results.

Post-task surveys measured perceived revision opportunities, judgment clarity, and cognitive load [11]. Participants were also asked to describe how they understood the role of the AI system during interaction, allowing exploration of how interface structure related to their interpretation of the generative process. Interaction logs recorded revision frequency, iteration count, and timing of judgment termination. Brief post-task interviews and think-aloud protocols captured participants’ explanations for stopping decisions.

3.3 Analysis

Survey responses were analyzed exploratorily using repeated-measures ANOVA to examine directional differences across interface conditions. Condition-level means and standard deviations are reported, along with test statistics and effect sizes. Behavioral logs were analyzed descriptively to characterize revision patterns and stopping behavior.

Qualitative data were open-coded to identify recurring judgment strategies, particularly distinctions between stopping due to uncertainty and stopping following perceived completion.

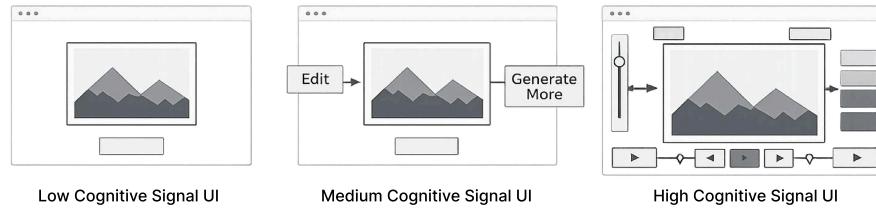


Figure 1: Wireframe overview of the three UI conditions by cognitive signal level (Low, Medium, High).

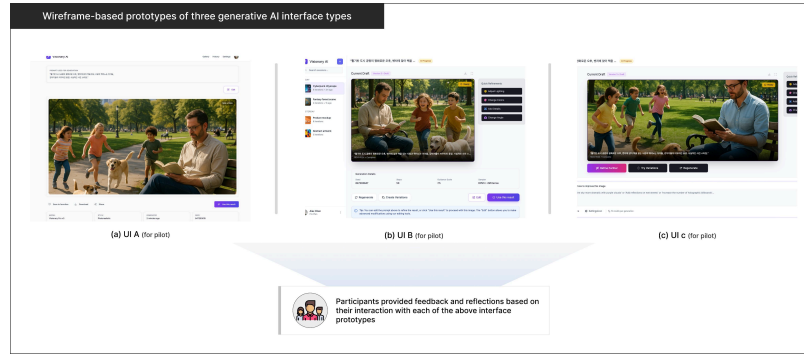


Figure 2: Interactive UI prototypes corresponding to each cognitive signal level.

4 Results

This section reports quantitative and qualitative differences in users' interpretations and stopping behaviors across UI conditions. Because identical AI-generated outputs were used in all conditions, observed differences can be attributed to interface structure.

4.1 Quantitative Results

4.1.1 Interpretation of AI Outputs Across UI Conditions. A repeated-measures ANOVA revealed a significant effect of UI condition on perceived judgment clarity, $F(2, 28) = 13.50$, $p < .001$, $\eta_p^2 = .49$. Mean clarity increased as cognitive signal levels increased (Low: $M = 2.53$, $SD = 1.13$; Medium: $M = 3.40$, $SD = 1.24$; High: $M = 4.40$, $SD = 0.74$).

In contrast, perceived editability did not significantly differ across conditions ($p > .05$), indicating that participants consistently recognized outputs as modifiable regardless of interface structure. Thus, differences were associated with when judgment terminated rather than whether modification was possible.

4.1.2 Modification and Judgment Termination Patterns. Behavioral patterns aligned with survey results. Under the Low Cognitive Signal UI, judgment frequently terminated early with limited revision. Under the High Cognitive Signal UI, participants more often engaged in iterative refinement before stopping.

Although effect sizes should be interpreted cautiously given the small sample ($N = 14$), the directional consistency across clarity ratings and revision behavior suggests that increasing the visibility and organization of editable points was associated with delayed

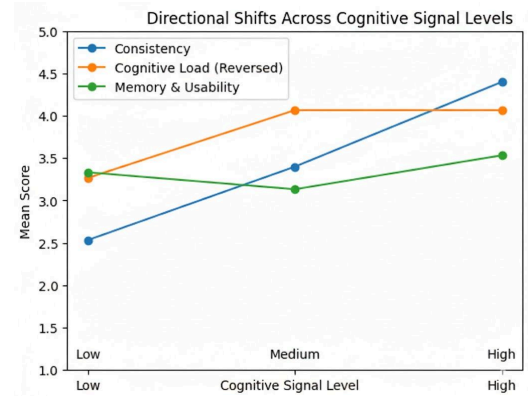


Figure 3: Overall UX profile across UI conditions by cognitive signal level. Error bars indicate ± 1 SD.

stopping and sustained interaction. Importantly, because the generated output remained constant, these differences reflect interface-level influences on judgment timing rather than output quality differences.

4.2 Qualitative Results

4.2.1 Stopping-from-Uncertainty vs. Stopping-from-Satisfaction. Qualitative analysis revealed two recurring stopping patterns. In the Low Cognitive Signal UI, participants often stopped due to uncertainty about next actions rather than satisfaction with the output. As one participant stated, "The result wasn't bad, but I couldn't

really see what I should do next, so it felt like I just stopped there” (P7).

In the High Cognitive Signal UI, stopping was more frequently described as a deliberate decision following iterative refinement. For example, one participant noted, “I could have kept tweaking it, but at some point I felt this was enough” (P5).

4.2.2 UI-Mediated Interpretation of Generative Outputs. Across interviews, participants did not differ in whether they believed AI outputs were editable, but in how strongly revision was framed as an expected next step. In generative AI contexts—where outputs are probabilistic and open-ended—interface cues appeared to shape whether outputs were interpreted as provisional or complete, thereby influencing when iterative engagement concluded.

5 Discussion

5.1 Cognitive Signals in Generative AI Interfaces and the Judgment Process

This study shows that interface-level cognitive signals were associated with differences in when users persisted, revised, or terminated judgment under identical AI outputs. Across UI conditions, judgments appeared to be guided less by output quality and more by cues about what actions were expected next. When cognitive signals were sparse, judgment tended to end early—not because users were satisfied—but because subsequent steps were difficult to organize. When cognitive signals were more clearly structured, users treated AI outputs as provisional units for evaluation and revision, integrating interpretation into an ongoing interaction rather than a terminal decision.

Unlike traditional editing tools, generative AI systems produce probabilistic outputs whose completion status is inherently ambiguous. In such contexts, users may lack stable mental models of how outputs can be steered or refined. Our findings suggest that interface structure plays a particularly important role in shaping judgment timing in generative AI workflows, where outputs are open-ended and iterative engagement is central to use. While similar affordance-related effects may occur in other systems, the provisional nature of AI-generated artifacts appears to amplify the influence of interface-level signals on when interaction concludes.

We do not position cognitive signals as a replacement for established affordance-based perspectives. Rather, we use the term to foreground how interface configurations organize not only action possibilities but the temporal dynamics of judgment—specifically, when users decide to continue or terminate interaction.

5.2 Design Implications for Generative AI Interfaces

The results suggest that in generative AI interface design, the presence of features may be less consequential than how those features are exposed and embedded within the interaction flow. The visibility and organization of editable points function as signals that implicitly shape expected next actions and the perceived scope of judgment, even without explicitly directing behavior.

First, generative AI interfaces may benefit from clearly structuring revision as part of the expected workflow rather than as an

optional afterthought. When modification features were not naturally integrated into the post-output phase, participants frequently experienced uncertainty and terminated judgment early. Second, interfaces may support both sustained iteration and appropriate stopping by making progression cues legible. In the high cognitive signal condition, participants described stopping not as interruption, but as a selective decision following sufficient exploration.

These findings highlight the importance of viewing generative AI systems not only as output generators but as interaction systems that mediate interpretation, action transition, and judgment termination.

This study is exploratory and based on a small within-subjects sample ($N = 14$). While the controlled design isolates interface effects from output variation, future work should examine diverse prompts, output qualities, and longer-term generative workflows to extend these findings.

6 Limitations and Future Work

This study examined generative AI interfaces within a controlled image-revision setting, and findings should be interpreted within this scope. The sample size was modest ($N = 14$), and although the within-subjects design supports internal comparison, results should be interpreted cautiously.

To isolate interface effects, the generated output was held constant. While this strengthens causal interpretation, it does not reflect the variability and prompt iteration typical of real-world generative AI use. Future work should examine how interface signals interact with diverse outputs and prompting strategies.

Participants primarily had prior design-tool experience, and judgment patterns may differ among less experienced users. We also did not directly measure mental models, trust, or authorship perceptions, which may interact with interface-level signals in shaping stopping decisions.

Finally, while this study focused on image generation, judgment dynamics may differ across text or code generation contexts, where notions of completeness and revision follow distinct interaction logics.

7 Conclusion

This study demonstrates that in generative AI image workflows, interface structure can influence how users experience the continuation and termination of judgment over AI-generated outputs. Examining interfaces through the lens of cognitive signals, we found that judgments were associated not only with output quality but with how editable points and subsequent actions were organized and made visible. Across identical AI outputs, participants interpreted results differently depending on interface configuration—treating them as provisional, adjustable, or complete. Interface structure did not change whether outputs were perceived as editable, but was associated with how revision unfolded and when judgment termination occurred. These findings underscore the importance of attending to interface-level signals when designing generative AI systems. Beyond supporting exploration, interfaces may shape how users recognize and approach meaningful stopping points within iterative interaction.

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